1. What are the three main methods to bulk insert data into SQL Server using SSIS?
The three main ways are:

- Using the **Bulk Insert Task**, which directly loads data from a file into a SQL Server table.
- Using the **Data Flow Task with an OLE DB Destination** set to "Fast Load" mode, which is probably the most common.
- Or, using the **SQL Server Destination** (though this one is less used and has some limitations, like it only works when SSIS runs on the same server as SQL Server).

---

2. In the File System Task, how do you configure SSIS to use a dynamic file path stored in a variable as the source?
You just go to the File System Task editor, set **IsSourcePathVariable** to True, and then pick the variable that holds your dynamic path. That way, instead of hardcoding a path, SSIS will grab the file location from the variable at runtime.

---

3. What SSIS component allows you to organize multiple tasks into a single logical group?
That's the **Sequence Container**. It basically acts like a box where you can put multiple tasks together and manage them as a unit.

---

4. In SSIS, what happens when you create two variables with the same name in different scopes?
Both variables exist, but SSIS will always use the one that's closest in scope. For example, if you have a package-level variable and another variable with the same name inside a Data Flow Task, the Data Flow Task will use its own version, not the package one.

---

5. Describe how the BatchSize and LastRow options influence the behavior of the SSIS Bulk Insert Task.

- **BatchSize** tells SSIS how many rows to send to SQL Server in one go. Using batches can help with performance and reduce memory usage.
- **LastRow** lets you stop the insert at a certain row. So, if your file has 10,000 rows and you set LastRow = 5,000, it will only insert the first 5,000.

---

6. Provide an example of how to use a Script Task in SSIS to read a flat file line by line.
Here's a simple example using C#:

```
using System.IO;
using Microsoft.SqlServer.Dts.Runtime;

public void Main()
{
   string filePath = @"C:\SSIS\input.txt";
   using (StreamReader reader = new StreamReader(filePath))
   {
      string line;
      while ((line = reader.ReadLine()) != null)
      {
         Dts.Events.FireInformation(0, "ReadLine", line, "", 0, ref bool fireAgain);
      }
   }
   Dts.TaskResult = (int)ScriptResults.Success;
}
```

This just reads the file line by line and writes each line to the SSIS log.

---

7. How do you remove the last comma from a list of values generated in a C# script in SSIS?

The simplest way is to use .TrimEnd(','). For example:

```
string values = "A,B,C,";
values = values.TrimEnd(',');
```

Now values becomes "A,B,C".

---

8. Explain how you would configure a File System Task to delete a specific file after it has been processed.

Set the **Operation** property to Delete File. Then, just like with copying or moving, set the

SourceConnection to point to your file (or to a variable if it's dynamic). Once the task runs, the file will be deleted.

---

9. What are the benefits of using a Sequence Container in large SSIS packages?
It makes your package more organized and easier to read. You can group related tasks together, control them as a unit (for example, disable the whole group or put a constraint on the group instead of each task), and even set variables or transactions at the container level. It really helps with managing large and complex packages.